



On the evolution of natural product biosynthesis

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Abstract

Natural products are the raw material for drug discovery programmes. Bioactive natural products are used extensively in medicine and agriculture and have found utility as antibiotics, immunosuppressives, anti-cancer drugs and anthelmintics. Remarkably, the natural role and what mechanisms drive evolution of these molecules is relatively poorly understood. The exponential increase in genome and

chemical data in recent years, coupled with technical advances in bioinformatics and genetics have enabled progress to be made in understanding the evolution of bio-synthetic gene clusters and the products of their enzymatic machinery. Here we discuss the diversity of natural products, incorporating the mechanisms that govern evolution of metabolic pathways and how this can be applied to biosynthetic gene clusters. We build on the nomenclature of natural products in terms of primary, integrated, secondary and specialised metabolism and place this within an ecology–evolutionary–developmental biology framework. This eco–evo–devo framework we believe will help to clarify the nature and use of the term specialised metabolites in the future.

“There is probably no more original, more complex, and bolder concept in the history of ideas than Darwin’s mechanistic explanation of adaptation.”

Ernst Mayr (1982)



1. Introduction

The evolution of metabolism enabled and shaped all the diversity of life on earth. The products of metabolism allow survival of life in virtually every environment on the planet, helping generate cellular energy, biomass and enabling communication and interaction between cells. The exploitation of metabolic products of life on earth by humans is testament to the versatility and diversity of their natural roles. Perhaps the best appreciated examples of metabolic products are the bioactive natural products (NPs), exemplified by those used in medicine and agriculture such as antibiotics, immunosuppressives, anti-cancer drugs and anthelmintics. In this review, we aim to capture why there may be such diversity in NPs, refine some of the terminology around metabolic products, review the current models for how NP pathways evolve, further develop the evolutionary framework for how new NP pathways emerge, placing these in the wider ecology–evolution–developmental biology architecture. This review will focus mostly on bacterial NPs, and often on those produced by bacteria belonging to the genus *Streptomyces*. There are multiple reasons for this—*Streptomyces* are amongst the most talented of organisms in terms of NP production. There are decades of research around *Streptomyces* and they are perhaps the best studied NP producers in terms of genomics, chemistry, and ecology and as such they provide the greatest amount of data on which to base the ideas developed in this discussion.



2. Diversity of natural products

The chemical diversity of small molecules produced by living organisms is exquisite. These NPs drive cellular life, inter- and intra-species interactions, communication and survival in every conceivable environment on Earth. Plants and microorganisms perhaps offer the best studied examples of bioactive NPs within the natural world, where the astounding diversity of chemistry produced by these organisms is far beyond the imagination and reach of synthetic chemists. In recent years the rate of discovery of novel NPs has accelerated exponentially due to technological advances and a realisation that the ability to produce NPs is important in a broad number of scientific fields beyond medicine and drug discovery including, but not limited to, ecology, evolution, biochemistry/biosynthesis and biotechnology.

It is estimated that more than 500,000 NPs have been documented in the scientific literature (Chassagne, Cabanac, Hubert, David, & Marti, 2019; Ntie-Kang & Svozil, 2020; Zeng et al., 2017), with actual numbers in nature likely to exceed this by several orders of magnitude. Remarkably, the diversity of NP chemistry possible is such that 83% of NP chemical scaffolds are non-existent in medicinal chemistry compound libraries (Cornell et al., 1995). Yet, given that the chemical space for compounds with a molecular weight <500 Da is likely to exceed 10^{60} , the number of bioactive molecules within this space represents a relatively small fraction of the possible chemospace (Boufridi & Quinn, 2016). Whilst approximately 70% of known NPs are produced by plants, microorganisms provide a significant number of the bioactive natural products employed in medicine and agriculture. Perhaps one of the best examples of NP discoveries was that of penicillin, which initiated the huge efforts to discover novel antimicrobial NPs from microorganisms. Despite being initiated by discovery of a NP in fungi, the so-called 'Golden age of antibiotic discovery' brought bacteria belonging to the taxon Actinobacteriota to the forefront of NP research. Indeed, one bacterial family within this phylum, the Streptomycetaceae provides one-third of currently categorised NPs of microbial origin (Bérdy, 2005; Chassagne et al., 2019). The Streptomycetaceae are abundant in every environment on Earth and the diversity of their NPs is likely driven by the necessity of competition, communication and interaction with other organisms (Clark et al., 2013; Meij, van der, Worsley, Hutchings, & Wezel, 2017; Seipke, Kaltenpoth, & Hutchings, 2012; Straight & Kolter, 2009). Bacteria are thought to account for around 9% of all NPs in chemical databases, with approximately 50% of these being bioactive molecules and around two-thirds of these have

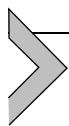
antimicrobial activities (Bérdy, 2005, 2012; Chassagne et al., 2019). Remarkably, many of the organisms that are talented producers of NPs also exhibit complex developmental and multicellular behaviours (sporulation, specialised cell types, symbiosis, predatory behaviours etc) suggesting that NP production may facilitate complex lifestyles and behaviours.

The remarkable diversity of chemical space in nature, is perhaps no less remarkable than the catalytic function and diversity of the enzymes that deliver the biosynthesis of these chemical entities. Over many years, NP producing microorganisms were selectively isolated to enrich for organisms with the ability to produce useful compounds. The emergence of affordable whole genome sequencing (WGS) and subsequent genome mining suggests that there are many more NP Biosynthetic Gene Clusters (BGCs) in the genomes microorganisms than our current experimentally validated inventory would suggest (Chevrette et al., 2021; Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson, & Barona-Gómez, 2019). For example, most environmental isolates of *Streptomyces* strains generally produce one or two NPs under laboratory conditions (Fernández-Martínez & Hoskisson, 2019). However, WGS reveals they often have the biosynthetic capability encoded in their genomes to produce far in excess of this number (Hoskisson & Seipke, 2020). Under laboratory conditions these so-called ‘silent BGCs’ are either not expressed or expressed at low levels (Hoskisson & Fernández-Martínez, 2018; Hoskisson & Seipke, 2020). One reason for this is that our understanding of the conditions and mechanisms that control NP biosynthesis are not fully understood and is limited to only a handful of well-understood molecules. Remarkably, these numbers are based on NPs where we have knowledge of their biosynthesis, enabling the use of genome mining tools and bioinformatics to identify them (Hoskisson & Seipke, 2020; Medema & Fischbach, 2015). There is likely also a large number of cryptic BGCs and products still to be discovered, representing a huge potential within chemical space (Hoskisson & Seipke, 2020). A deeper understanding of the evolution, biosynthesis and regulation of these NPs would enable us to rationally initiate biosynthesis of the ‘silent majority’ of BGCs, identify cryptic clusters and these could represent a vast source of new NPs (Fernández-Martínez & Hoskisson, 2019; Hoskisson & Fernández-Martínez, 2018; Hoskisson & Seipke, 2020).

The evolution of enzymes in NP biosynthetic pathways is likely subject to the same kinds of evolutionary selection mechanisms as other enzymes in biochemical pathways (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019). Increasing evidence however indicates that some of

this diversity is not driven by highly adapted, specialised enzymes, but through enzyme promiscuity—the ability of enzymes to act on diverse substrates, often beyond the substrates that have shaped their prior evolution (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019; Khersonsky & Tawfik, 2010; Peracchi, 2018). Whilst ‘one-gene, one reaction’ enzymology may contribute to the biosynthesis of a complex NP, the complexity of many NPs requires whole clusters of genes to deliver the final NP. These BGCs have been considered as ‘gene collectives’, where natural selection can act to deliver innovation and chemical diversity (Fischbach, Walsh, & Clardy, 2008), where one pathway can deliver many products (Fischbach & Clardy, 2007). In some cases, detailed phylogenetic analyses of NP biosynthetic pathways, such as the evolution complex megasynthase enzymes, have helped understand the genetic basis of pathway and product diversity (Grininger, 2023; Jenke-Kodama, Sandmann, Müller, & Dittmann, 2005). More recently, holistic approaches to reconcile chemical, functional and genetic data within evolutionary models have emerged, such as the dynamic chemical matrix evolutionary hypothesis, which enables the relationships between chemical distance, bimolecular activity, selection and adaptation to be considered in the context of a fitness landscape (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019).

These models help, in part, to describe what is observed in nature, but understanding the mechanisms that drive the emergence of novel biosynthetic pathways requires us to consider the wider ecological and evolutionary (eco-evo) context of NP production. The availability of large genomic and chemical datasets, coupled with increasingly sophisticated bioinformatics is now giving us access to an increasing diversity of genetically related BGCs and the molecules they produce, enabling a greater understanding of the evolutionary forces that shape the emergence of new NPs (Chevrette et al., 2021). Coupling these data with greater knowledge of the ecology, physiology and developmental biology of producing organisms will enable greater insight into the forces that shape production of bioactive NPs. After all, the diversity of NPs in nature is the result of evolutionary sampling at the interface of chemical and biological space.



3. Primary, secondary and specialised metabolism

Conventional thought considers primary metabolism as those pathways that normally give rise to a single product, and that the products of

these pathways are utilised for ‘usual’ cellular functions (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019; Fernández-Martínez & Hoskisson, 2019; Firm & Jones, 2009; Fischbach & Clardy, 2007). This core set of metabolic reactions also appears to be evolutionary conserved across many organisms (Peregrín-Alvarez, Sanford, & Parkinson, 2009) with the metabolites in these pathways numbering only several hundred compounds (Fernández-Martínez & Hoskisson, 2019). Moreover, the enzymes that comprise these pathways are thought to have been subject to prolonged periods of natural selection resulting in highly adapted efficient catalytic mechanisms with high end-product yields (Firm & Jones, 2009). Another feature of primary metabolic pathways is multi-level regulation, such that primary metabolism can be regulated at the transcriptional, post-transcriptional, translation and post-translational level (Berkhout, Teusink, & Bruggeman, 2013; McAdams, Srinivasan, & Arkin, 2004; Seshasayee, Bertone, Fraser, & Luscombe, 2006; Soyer & Pfeiffer, 2010), or some combination all these mechanisms. This multilevel control helps to provide high levels of robustness and responsiveness to environmental perturbation and fluctuations (Berkhout et al., 2013; McAdams et al., 2004; Seshasayee et al., 2006; Soyer & Pfeiffer, 2010). The wide taxonomic distribution of these pathways and the relative lack of diversity in primary metabolic pathways across organisms (reflecting the shared nature of primary metabolic intermediates) led Firm and Jones (2009) to suggest the use of the term ‘integrated metabolism’ rather than primary metabolism. However, ‘integrated metabolism’ as a term may add further confusion to the issue, as it suggests that there is limited or no cross-over between primary and secondary metabolism—the building blocks for NP biosynthesis are derived from primary metabolism, thus intimately integrating both forms of metabolism (Schniete et al., 2018).

Secondary metabolism is a term coined more than a century ago (Kössel, 1891). It was developed to reflect the occurrence of metabolites that are produced from distinct, taxonomically restricted pathways that are, by implication, less important to an organism—being secondary to normal cellular processes (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019; Fernández-Martínez & Hoskisson, 2019; Firm & Jones, 2009; Kössel, 1891). These metabolites are frequently the bioactive NPs discussed in the previous section. Secondary metabolism is, however, a misleading term, because many of these compounds play significant roles in the biology of the producing organism [for example siderophores (Kramer, Özkaya, & Kümmerli, 2019), quorum sensing molecules (Daer et al., 2018;

Kudo et al., 2020), germination inhibitory metabolites and spore coating NPs (Aoki, Matsumoto, Kawaide, & Natsume, 2011; Sigle, Ladwig, Wohlleben, & Muth, 2015), bioactive lipids (Murata, Sugiyama, Matsuoka, & Matsumori, 2015), carotenoids (Li, Swofford, & Sinskey, 2020)]. Moreover, cultivation and evolution of strains in the laboratory under artificial conditions is not representative of the natural niche occupied by these organisms and the coordinated regulation of their biosynthesis may result in altered production profiles in the laboratory and artefacts of growing strains at high density (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019).

To address this conflict in terminology the expression 'specialised metabolism' entered the arena. It is often used interchangeably with secondary metabolism, further confusing the field. First coined by Davies (2013) in the context of microorganisms to take into account the ecological role played by some NPs in the producing organisms, emphasising that these NPs are in no sense secondary (Price-Whelan, Dietrich, & Newman, 2006). The term has been widely adopted, but often used inappropriately due to a lack of knowledge on how they are 'specialised'.

We recently argued that the definition of these three types of metabolism (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019) be formalised to consider the ecology of the producer and to recognise that in order to refer to these molecules as specialised, without knowing their ecological role, is perhaps premature. Secondary metabolism, as earlier definitions suggest, is less integrated and with a more limited taxonomic distribution (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019), and where the ecological role of the product is currently unknown. We also argued that these pathways exhibit high degrees of enzyme promiscuity and co-option (enzyme recruitment) than is observed in primary metabolism and this may drive the emergence of 'side activities' upon which selection can act (Barrick & Lenski, 2013; Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019; Guzmán et al., 2015; Khersonsky & Tawfik, 2010). We postulated that specialised metabolism reflects a distinct area of secondary metabolism, where the role of the NP is understood in an ecological context, whilst recognising that experimentally validated niche-specific roles are rare. Where we do have some knowledge of the role these NPs play in the biology and ecology of the producing organism, we can confidently refer to them as specialised metabolites.

Here we wish to expand on these definitions beyond just consideration of the ecological context of NPs and consider these in terms of their

ecology, evolutionary and developmental biology (eco–evo–devo). In doing this, it will add further considerations to whether a NP is considered a specialised metabolite. The development of ‘specialised metabolism’ as a concept first emerged in the plant field, where tissue specific expression, ecological interaction and stress response could be considered for NPs (Firn & Jones, 2009). In the field of bacterial NPs, many of the producing strain have remarkable niche adaptations (Fahimipour & Gross, 2020) and have evolved complex developmental lifecycles, multicellular growth forms and the formation of resting/survival strategies (Claessen, Rozen, Kuipers, Søgaard-Andersen, & Wezel, 2014), perhaps analogous to the root, stem, leaves and seeds of plants that inspired Firn and Jones. Although genome sequence data can inform on some aspects of these, there is a great need for experimental data to augment NP studies.



4. Evolution of metabolic pathways

The emergence of interconnected metabolic pathways represents a significant step forward in the evolution of cellular life (Fani & Fondi, 2009). How these complex metabolic networks form in genomes, through the clustering of genes has been an area of great interest across all domains of life for many years (Horowitz, 1945; Horowitz, 1965). The combination of biochemical data with the emergence of genomic data has led to important insight into how pathways have evolved via the convergence of different reactions/pathways, such as the joining of the oxidative branch of the citric acid cycle with the reductive branch of the cycle (Huynen et al., 1999; Meléndez-Hevia, Waddell, & Cascante, 1996).

There are several proposed mechanisms that are believed to have helped shape genome evolution and the emergence of metabolic pathways (Schmidt, Sunyaev, Bork, & Dandekar, 2003). These have primarily been considered in terms of adoption/recruitment, gene duplication and expansion via HGT, gene fusion, and HGT of novel genes, tandem duplications within coding sequences, non-orthologous gene displacement (Bolotin, Melamed, & Livnat, 2023; Fani & Fondi, 2009; Fani, Brilli, Fondi, & Lió, 2007; Fondi, Emiliani, & Fani, 2009; Forterre, 1999; Näsval, Sun, Roth, & Andersson, 2012; Noda-García & Barona-Gómez, 2013; Schmidt et al., 2003; Scossa & Fernie, 2020; Snel, Bork, & Huynen, 2000; Figs. 1 and 3). The development of ideas such as the ‘Patchwork Hypothesis’ that postulates that recruitment of ancestral enzymes, that can

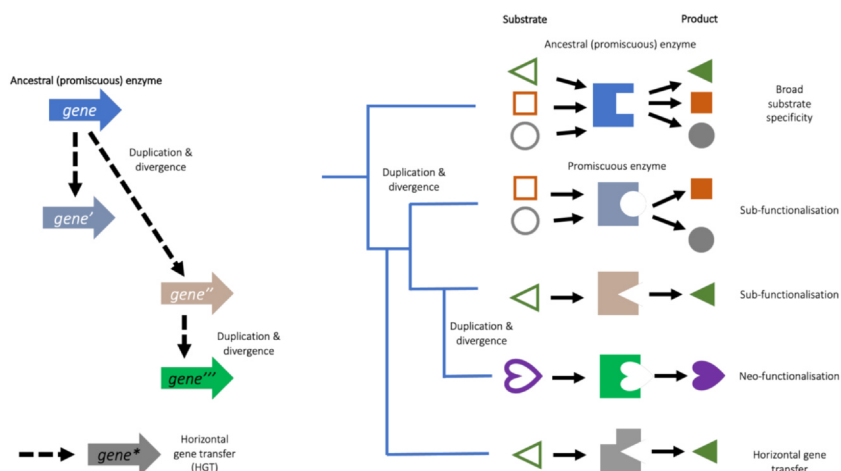


Fig. 1 Gene expansion events and evolution of gene function. Duplication of genes can lead to divergence of function in the duplicate copies (*gene'*, *gene''*, *gene'''*) along with gene expansion through horizontal gene transfer (HGT; *gene**). The ancestral gene (*gene*) may have promiscuous function in terms of reactions and substrates accepted. Duplication can lead to the streamlining of this functionality as genes adapt to new roles without the evolutionary constraint of an ancestral function, while still potentially accepting the same substrates as the ancestor (subfunctionalisation). Neofunctionalisation occurs when enzyme function diverges sufficiently (*gene'''*) to create novel activities with a substrate no longer accepted by the ancestral gene. Horizontally transferred genes can have overlapping substrates, have different phylogenetic signals within trees, but may still have overlapping enzyme function with genes already present within an organism. These events can be tracked phylogenetically. It is likely that HGT occurs more frequently than duplication (Treangen et al., 2011), with promiscuity in function being key to evolutionary sampling.

react with a wide range of substrates, can facilitate emergence of pathways that are further shaped by adaptive evolution (Fondi et al., 2009; Jensen, 1976; Yčas, 1974). In this model, duplication of genes can lead to substrate/cofactor specificity evolving, facilitating divergence and novel functionality (Fig. 1). Gene duplication has an important prerequisite for gene innovation through the early evolution of life on Earth (Copley, 2020) and is thought to have enabled duplicates to become specialised. While in these models, the most likely fate of a redundant duplicate is non-functionalisation, there is potential for innovation. Models explaining neofunctionalisation and subfunctionalisation (Lynch & Force, 2000) have frequently been used to explain the retention of duplicate functionality (Fig. 1). Neofunctionalisation suggests that the gain of new functions is the



Fig. 2 Architecture of a natural product producing biosynthetic gene cluster. This is the BGC for the biosynthesis of aurodox, a hybrid PKS/NRPS antibiotic from *Streptomyces goldiniensis* (McHugh et al., 2022). Many NP BGCs have conserved features. The presence of structural genes for the generation of the NP scaffold, genes for tailoring the NP, modifying the final product, regulatory genes (cluster situated regulators), often a transport mechanism (for secreted products) and in some cases a cluster specific immunity mechanism. These immunity/resistance genes can often be located elsewhere on the chromosome.

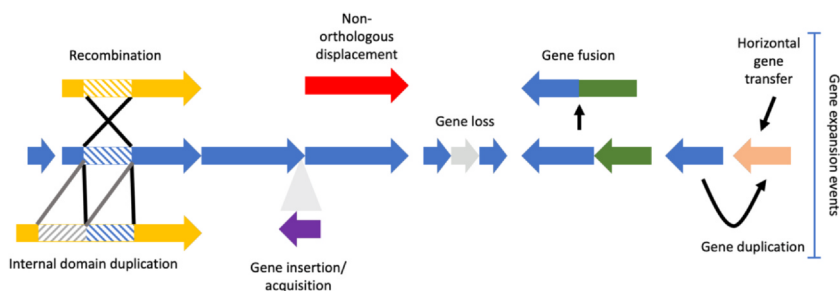


Fig. 3 Evolutionary mechanisms occurring within biosynthetic gene clusters. Several mechanisms affect the evolution of BGCs. Recombination between homologous genes and domains with and beyond BGCs can facilitate innovation within clusters. Tandem duplications within genes can occur to diversify function, such as duplication of modules in PKS/NRPS enzymes, expanding cluster and product diversity. Genes may be inserted/acquired or lost from BGCs, or can be replaced through non-homologous gene displacement. These processes can change the final product of the BGC, such that novel activities can be sampled, changes to product stability, uptake, secretion etc can be achieved. Genes can fuse to generating novel enzyme activities. Gene expansion events (gene duplication and HGT of orthologs) can lead to sub- and neofunctionalisation within BGCs leading to clusters of genes producing novel products. The promiscuity of enzymes, in terms of substrates accepted in these expansion events, is a key feature of the evolution of BGCs.

major selective factor for the retention of both duplicate genes within a genome. While subfunctionalisation suggests that both duplicate genes undergo mutations, such that both copies are required to fully complement the ancestral gene (likely non-adaptive). It is also possible for both mechanisms to work in parallel, with rapid subfunctionalisation following the duplication event, with subsequent neofunctionalisation occurring

(He & Zhang, 2005). This model is also consistent with the idea that ancestral enzymes can catalyse similar reactions on similar substrates, such that metabolic pathways can expand their capabilities, leading to the emergence of novel enzyme activities and pathways (Brilli & Fani, 2004). Remarkably, there are examples where promiscuity is maintained within related pathways and where HGT has contributed to subsequent pathway evolution (Barona-Gómez & Hodgson, 2003; Glasner, Truong, & Morse, 2019; Noda-García & Barona-Gómez, 2013; Noda-García et al., 2013; Fig. 1).

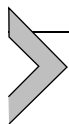
Recruitment of existing enzymes into metabolic pathways is thought to have occurred frequently, with examples such as tRNA synthetases being prominent (Krahn, Söll, & Vargas-Rodriguez, 2022). It is thought that tRNA synthetases evolved independently and were then likely recruited into the wider functions we see today such as in translation and tRNA-dependent transamination (Krahn et al., 2022; Min, Pelaschier, Graham, Tumbula-Hansen, & Söll, 2002). The increase in genome data also led to the hypothesis that many genes in bacterial genomes were the result of gene duplication events (Rosa & Labedan, 1998). However, as the datasets became larger and bioinformatic tools became more sophisticated, it was apparent that HGT not duplication was driving protein family expansion in prokaryotes (Treangen, Rocha, & Moran, 2011). Coupling the processes of HGT and gene duplication together (gene expansion), is believed to be the most widely accepted model for the emergence and evolution of novel metabolic pathways (Fig. 1). There is still much work to be done in understanding the frequency of both processes in different environmental niches.

In bacterial genomes around 50% of genes are organised as operons (Nützmann, Scazzocchio, & Osbourn, 2018). There are several models that have been proposed for the emergence of operon structure (Fondi et al., 2009). Often these overlap with some of the hypotheses around the emergence of metabolic pathways, such as duplication and divergence leading to 'assembly lines' of genes (Horowitz, 1945), other models suggest that proximity of alleles contributing to a pathway help to restrict unwanted homologous recombination could favour operon formation (Fondi et al., 2009). Incorporation of additional biological features into models that could favour operon formation such as the transcriptional-translational coupling that is observed in bacteria (Glansdorff, 1999) and/or that co-regulation is favourable under fewer transcription factors (TFs)/promoters/regulation sites (Lawrence & Roth, 1996; Lawrence, 1999). These models are discussed in great detail by Fondi et al. (2009), with many

having overlapping features both with each other and with the pathway evolution models. This suggests that many of these forces could be at play under different selection/adaptation scenarios.

Correct functioning of new pathways not only relies on appropriate processing/biosynthetic enzyme recruitment and catalytic activity but also on the regulatory mechanisms that can adjust expression of genes and enzymes to ensure biosynthesis can proceed. The majority of regulatory signals emerge from changes in metabolic intermediates (Ledezma-Tejeda, Schastnaya, & Sauer, 2021; West, Delattre, Noor, Feliu, & Soyer, 2023) and likely result in conservation of regulatory networks for metabolism. In related species, orthologous regulators control cellular responses to signals, and yet there is exceptional plasticity in the regulatory mechanisms and organisation of operons (McAdams et al., 2004; Perez & Groisman, 2009a). The movement of genes within and between genomes by homologous recombination and HGT can also facilitate the emergence of new regulatory processes, all against a background of the mutational rate (point mutations) of a given organism (Babu, Teichmann, & Aravind, 2006; Chen, Jewett, & Groisman, 2011; Choi & Groisman, 2020; McAdams et al., 2004; Perez & Groisman, 2009a, 2009b). Remarkably, the acquisition of regulatory function appears to be similar in mechanism to how structural genes are recruited to form pathways, however with the opportunity to regulate at a number of different levels within the cell, it offers great opportunity for adaptation to occur in spatial, temporal, genetic and chemical contexts (McAdams et al., 2004). The evolution and fixation of mechanisms within operons that regulate metabolic pathways at the post-transcriptional and post-translational level is relatively poorly studied (Bradley, 2022; Naville & Gautheret, 2010). Post-transcriptional and post-translational level control of genes and enzymes is pervasive in nature and the complexity of these control mechanisms scales with organismal complexity (Bradley, 2022). It is clear that adding fine level regulation to metabolism extends the controllability of pathways (Kochanowski, Sauer, & Noor, 2015), and can help fix adaptive functions (Brunk et al., 2017), but we much still to learn about the mechanisms of how they evolve and become fixed in genomes. It does however appear that duplication and HGT also play a role in the emergence of TFs (Barrick & Breaker, 2007; Clark & Hoskisson, 2011; Naville & Gautheret, 2010; Vitreschak, Mironov, Lyubetsky, & Gelfand, 2008). Moreover, experimental evolution studies indicate that evolution of control regions for genes can happen rapidly when a selective pressure to use particular

gene is applied (Dabizzi, Ammannato, & Fani, 2001; Kasak, Hôrak, & Kivisaar, 1997) and that transcription factors evolve faster than their targets (Wheeler et al., 2022).



5. Architecture of biosynthetic gene clusters

The genes encoding the enzymes for the biosynthesis of NPs in bacteria and fungi are generally found contiguously in the genomes of producing organisms (Keller, 2019; Medema & Fischbach, 2015; Tran, Yen, Chiang, Lin, & Chen, 2019). There are of course several caveats to this—often for practical or technical reasons—some BGCs are not able to be cloned due to the lack of resistance genes in the BGC, which may be located elsewhere on the chromosome (see below), many BGCs are identified through bioactivity, those with narrow activity ranges can be lost during screening processes, and more recently, we rely on bioinformatic genome mining to find BGCs that look like what we already know. There may be a whole wealth of NP biosynthetic genes that differ from the canonical view of the BGC.

The clustering of genes in BGCs in bacteria is unsurprising as in bacteria almost half of the genes in each genome are organised as operons (Nützmann et al., 2018). Conversely, in plants the clustering of NP biosynthetic genes is limited, although there are some examples of clustered genes within primary and secondary metabolic pathways (Nützmann et al., 2018; Osbourn, 2010a). The diversity of NPs produced by bacteria is reflected in the diversity of BGCs that are found in nature. There are however, a number of common themes within these BGCs that facilitate biosynthesis of such a diverse range of chemical entities. In general the genes for the scaffold of the NP are present within the BGC, the enzymes for tailoring their biosynthesis, genes that encode one or more transcription factors (TFs) to regulate biosynthesis, genes that facilitate transport of the NP following biosynthesis and an immunity mechanism (Bibb, 2005; Blin et al., 2021; Medema & Fischbach, 2015; Navarro-Muñoz et al., 2019; Terlouw et al., 2022; Wu, Zhang, Deng, & Yu, 2022).

There are, of course, variations on these themes—transport and immunity mechanisms may be encoded by the same genes (Crits-Christoph, Bhattacharya, Olm, Song, & Banfield, 2020; Severi & Thomas, 2019). TFs and regulatory proteins may originate from other BGCs (McLean, Hoskisson, & Seipke, 2016; Santamarta, Rodriguez-Garcia,

Perez-Redondo, Martin, & Liras, 2002) and may fully or in part co-regulate more than one BGC (Hoskisson & Fernández-Martínez, 2018). For many years the concept of a pathway specific regulator was pervasive within the field, new examples are emerging all the time that illustrate variations in regulation of NP biosynthesis, such as pathway specific sigma factors, pathway specific two-component systems and cross-pathway regulators (Hoskisson & Fernández-Martínez, 2018). There are also increasing numbers of examples of enzymes recruited from primary metabolism, that act upon primary metabolic substrates to deliver or modify precursors for NPs (Sélem-Mojica, Aguilar, Gutiérrez-García, Martínez-Guerrero, & Barona-Gómez, 2019) and even examples of NPs which are the biosynthetic precursors of other NPs (Qin et al., 2020) or the pathways bifurcate (Booth et al., 2022).

The schematic (Fig. 2) shows the organisation of a canonical BGC, where the scaffold generating enzymes (in this case (McHugh et al., 2022), a series of large megasynthases encoding polyketide synthases [PKS] and non-ribosomal peptide synthases [NRPS]). The tailoring enzymes are clustered and consist of a range of enzyme families and within these are collocated the regulatory genes and the resistance and transport mechanism. Contained within BGCs are genes that have been acquired from primary metabolism which has led to a number of hypotheses around how these can give rise to novel activities through expansion (duplication or Horizontal Gene Transfer [HGT]) and recruitment to the cluster, where enzymes have relaxed constraints of delivering their primary metabolic function may evolve novel substrate specificities and/or activities (Booth et al., 2022; Chevette, Gutiérrez-García et al., 2019; Chevette, Hoskisson et al., 2019). Remarkably, in many BGCs recruited genes may still resemble their primary metabolic counterparts (e.g aspartate decarboxylates (McHugh et al., 2022; Weber et al., 2008), 3-carboxyvinyl-phosphoshikimate transferase (Cruz-Morales et al., 2016), glutamate dehydrogenase (Sélem-Mojica et al., 2019), dehydroquinase synthase (Sélem-Mojica et al., 2019)).

Resistance mechanisms encoded within BGCs are increasingly under scrutiny in terms of how essential they are for the formation of a BGC (Wu et al., 2022). While conventional thought suggests that export/resistance function is required within the BGC (Liu, Hao, Xie, Horsman, & Chen, 2016; Martín, Casqueiro, & Liras, 2005; Severi & Thomas, 2019), there are increasing examples of biosynthesis and self-resistance overlapping (Cui et al. 2018; Cui et al., 2020; Wenciewicz, 2019; Wu et al., 2022). It is also being observed with increasing regularity that resistance to antimicrobial

NPs can be achieved through duplication of the cellular target in producers to provide resistance, and often not colocated with the BGC (Alanjary et al., 2017; Kale, McGlinchey, Lechner, & Moore, 2011; Tang et al., 2015).

The conserved nature of BGCs—clustering/co-localisation of genes, presence of resistance mechanisms—has enabled a range of bioinformatic tools to be developed that use a range of these and other features to identify BGCs such as antiSMASH (Blin et al., 2021), GECCO (Carroll et al., 2021), plantiSMASH (Kautsar, Suarez Duran, Blin, Osbourn, & Medema, 2017), ARTS (Alanjary et al., 2017), PRISM (Skinnider et al., 2020) and other tools for genome mining NPs (Hannigan et al., 2019; Skinnider et al., 2016; Walker & Clardy, 2021). These tools have further lead to evolutionary inspired methods search and identify novel NP clusters (Cruz-Morales et al., 2016; Kautsar, van der Hooft, de Ridder, & Medema, 2021; Navarro-Muñoz et al., 2019; Sélem-Mojica et al., 2019).



6. Evolutionary models for emergence of new natural product biosynthetic clusters

Current models for the evolution of NP BGCs are largely based on mechanisms that facilitate the emergence of primary metabolic pathways. There are however fundamental differences when one considers how selection may act upon these pathways. The reasoning is that rather than conferring a growth phenotype, NPs confer a survival benefit to the producing organism by binding cellular targets in competitors (Stone & Williams, 1992; Williams, Stone, Hauck, & Rahman, 1989). There are numerous validated targets for antibacterial NPs that target highly conserved cellular processes maintained in bacteria, for example Lipid II (Breukink & Kruijff, 2006), but there are numerous others to be found in the literature that act on cell wall synthesis alone (see (Sarkar, Yarlagaadda, Ghosh, & Haldar, 2017) and references therein) and the bacterial ribosome (Hansen, Moore, & Steitz, 2003). Fischbach and Clardy (Fischbach & Clardy, 2007) described this as a ‘target-based’ model of evolution. Experimental attempts have also been proposed and tried to evolve producing strains to overcome resistant competitor organisms in this target-based manner (Charusanti et al., 2012; Waldetoft, Gurney, Lachance, Hoskisson, & Brown, 2019). The driving force behind the ‘target-based’ model is the exquisite fit of NP with the cellular target (Fischbach & Clardy, 2007). There are of course caveats to this as a valid selective

pressure on NPs. The molecules where modes of action are well studied are often not natural targets, for example, there are excellent binding kinetics of some molecules with cellular targets that they may never have seen during their evolution—indeed, the example of rapamycin binding to mTOR-rap-FKBP given by Fischbach and Clardy (Fischbach & Clardy, 2007) is likely one such example.

The models proposed by Firm and Jones (Firm & Jones, 2000, 2009) suggested a different view of NP evolution, largely developed on the observation that few biological molecules have potent bioactivity (in areas we know how to screen for), and indeed known bioactivity in chemospace is limited (Boufridi & Quinn, 2016). Given the energetic cost to synthesise a range of molecules, Firm and Jones (2000, 2009) postulated that organisms that can synthesise chemical diversity at low energetic cost would enable screening of a wide range of NPs, where beneficial bioactivity could subsequently be fixed in a population. Evidence supporting this, such as natural products being produced as a suite of compounds and the branching of pathways, is growing in the literature (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019; Fischbach & Clardy, 2007).

A major challenge in natural product biosynthesis and chemistry is to develop a more satisfying explanation for how NP BGCs evolve, placing these within a robust evolutionary framework. Progress in this area was made by Chevrette, Gutiérrez-García et al. (2019), Chevrette, Hoskisson et al. (2019) with the development of the dynamic chemical matrix evolutionary (DCME) hypothesis, where chemical distance, bimolecular activity and organismal fitness shape the adaptive landscape of novel NP emergence. The DCME hypothesis and the ‘screening hypothesis’ of Firm and Jones (2009, 2000) account for the evolutionary forces acting upon NP biosynthesis and the evolution of their BGCs (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019). As one would expect the enzymes within BGCs (or proto-BGCs) would serve as the main diversifying forces for delivering novel NPs. The discussions above have shown how the duplication, expansion, gene fusion, recruitment and promiscuity can aid the formation of operons (clustering of genes and the emergence of co-regulation). Given the increasing evidence for genomes being highly dynamic, in terms of their arrangement and structure, than previously thought (Brussow, Canchaya, & Hardt, 2004; Page, Ainsworth, & Langridge, 2020), which can facilitate the evolutionary sampling of the ‘gene collectives’ idea of Fischbach and Clardy (2007). The ‘screening’

proposed by [Firn and Jones \(2009, 2000\)](#) is likely driven by purifying and positive selection acting at these loci. The DCME hypothesis proposes that in the early stages of evolution, the cluster is largely free from selection (neutral evolution), consistent with the ‘sampling hypothesis’ and with the original ideas of [Kimura \(1983\)](#). However, a note of caution is perhaps required here, the large effective population sizes of bacteria and the coding density within genomes, would likely mean that most mutations in bacterial genomes are deleterious ([Rocha, 2018](#)). This suggests that the supply of mutations (recruitment events, duplications, HGT, etc.) needs to have a high rate in order to deliver sufficient ‘sampling’ events to enable fixing within genomes of NP producing bacteria. To develop a better understanding of this aspect of the model, we require more data in the area around understanding mutation rates and HGT of bacteria in the natural environment ([Rocha, 2018](#)). The limited data available in the context of adaptive mutations being fixed suggests that in enteric bacteria around 50% of non-synonymous mutations become fixed through positive selection ([Charlesworth & Eyre-Walker, 2006](#)). There is also emerging data that suggests that homologous recombination in NP producing soil *Streptomyces* is much greater than was previously appreciated and that this scales with phylogenetic distance ([Doroghazi & Buckley, 2010](#); [McDonald & Currie, 2017](#); [Tidjani et al., 2019](#)). This suggests that recruitment could be at a higher rate than previously appreciated. This would provide the high levels of mutation for evolution to sample and potentially fix newly formed clusters within genomes if they are adaptive. Moreover, the DCME hypothesis suggests positive selection this would be increased concomitantly with chemical diversity.

One important aspect of the DCME hypothesis is the incorporation of purifying selection in the model. While genome flexibility (facilitating the high mutation rate) enables rapid responses to environmental change and ecological adaptation through pangenomic variation in microorganisms ([Lee & Marx, 2012](#); [McInerney, McNally, & O’Connell, 2017](#); [Niehus, Mitri, Fletcher, & Foster, 2015](#); [Porter, Chang, Conow, Dunham, & Friesen, 2017](#)) experimental evidence for purifying selection in the environment is now beginning to emerge and appears to be largely driven by environmental stress ([Simonsen, 2022](#)).

In the context of NP production and the DCME hypothesis, understanding what is driving the types of selection proposed in the model is a challenge. There has been a general assumption in the NP research community that the production of a novel molecule is likely to become fixed

and subject to positive selection rapidly (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019), yet this is contrary to the evidence that genomic changes in bacteria are generally deleterious (Rocha, 2018). Purifying selection could be acting for a number of reasons in terms of how producing organisms respond to the birth of a new cluster, such as the emergence of a novel BGC/NP could be energetically unfavourable. For example, many NPs consume large amounts of reducing power in biosynthesis and this can affect other aspects of cellular efficiency (Barra et al., 2021; Chemler, Fowler, McHugh, & Koffas, 2010). Likewise, ATP consumption during biosynthesis or during NP export could impact on the energy poise within cells (Hara & Kondo, 2015), again resulting in deleterious effects. The requirement for immunity/resistance to novel natural products could also represent a significant purifying selection pressure on the sampling of chemical space by new gene collectives. While surrogate efflux pumps are known to confer resistance to a range of antimicrobial NPs (Chitsaz & Brown, 2017) and many of these are found as resistance mechanisms within BGCs (Cris-Christoph et al., 2020; Severi & Thomas, 2019). The birth of new clusters, with potentially toxic bioactivities of novel NPs will require these strategies. This can be solved in part by promiscuous drug exporters (Chitsaz & Brown, 2017), but also by duplication of cellular targets (the premise that ARTS is based upon identifying (Alanjary et al., 2017)). It has also been shown that antibiotic heteroresistance can be achieved rapidly by spontaneous gene amplification events (Nicoloff, Hjort, Levin, & Andersson, 2019). These spontaneous and unstable heteroresistant phenotypes also rapidly revert in the absence of the antibiotic. This offers an intriguing evolutionary solution to rapidly evolve resistance to a novel NP and is consistent with immunity mechanisms laying outside of the BGC (Thaker et al. 2013; Thaker, Waglechner, & Wright, 2014) and allows for a mechanism of reversion should the cluster not become fixed. The mechanism thus enables recruitment, sampling and resistance to be subjected to natural selection.

Reconciling these models with NP pathways and BGCs that are better understood in terms of their evolution, supports many of these features the evolution of PKS and NRPS containing NP BGCs have evidence to support natural hybridisation of pathways (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019) and gene fusions (Coleman, Hogan, McGoldrick, Rudden, & McNerney, 2015). Homologous recombination within these large repetitive megasynthases is likely frequent and shown to generate diversity not only between species, but also within

species and between BGCs of the same enzyme family (Callahan, Thattai, & Shraiman, 2009; Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019; Grninger, 2023; Jenke-Kodama, Börner, & Dittmann, 2006; Nivina, Yuet, Hsu, & Khosla, 2019; Ridley, Lee, & Khosla, 2008). More widely hybridisation of pathways can occur and Evomining of genomes has identified further examples of pathway hybridisation (Navarro-Muñoz et al., 2019) which can help in expanding chemical diversity in a pathway rapidly.



7. Reconciling the constraints of current BGC evolution models

The BGCs for specialised metabolites are complex multi-part traits—each enzyme contributes a precursor or modifies/decorates a chemical scaffold to deliver a bioactive metabolite (or more) with a specific role either in the producing organism, or that can act upon another organism. The evolution of multi-part traits has puzzled biologists since Darwin, in terms of how apparently highly integrated, functionally complex structures and systems can originate through evolutionary processes. The evolutionary processes that lead to the complexity and innovation as different traits come together to deliver function. The incremental progressions of pathways may at first always seem like they have to proceed in a linear manner from simple to complex, however gene loss as well as gain within clusters is equally likely [e.g. A47934 & Teicoplanin BGCs (Donadio, Sosio, Stegmann, Weber, & Wohlleben, 2005), lantibiotics [RiPPs] (Dufour, Hindré, Haras, & Pennec, 2007)]. Selection acts upon phenotypes not genotypes, therefore there needs to be significant standing genetic variation within a population for selection to act upon, therefore population genetics and dynamics of producing organisms needs to be considered fully to understand evolution of BGCs, currently we know little about these processes although progress has been made recently (Doroghazi & Buckley, 2010; Tidjani et al., 2019).



8. Considerations for expanding the evolutionary models of biosynthetic gene clusters

There are four key questions to be considered when reconciling BGC evolutionary models (Fig. 4).

- 1) **What types of mutations are required to deliver innovation in biosynthetic gene clusters?** Fixation of mutations in a population is the primary source of evolutionary change in biology, the discussion above indicates that recruitment, gene duplication, gene fusion and HGT all play a role in BGC evolution (Figs. 1 and 3; Bolotin et al., 2023; Fani & Fondi, 2009; Fani et al., 2007; Fondi et al., 2009; Näsvall et al., 2012; Noda-García & Barona-Gómez, 2013; Schmidt et al., 2003; Scossa & Fernie, 2020; Snel et al., 2000), with genetic drift (point mutations) continuing to provide variation for evolution to act upon. In addition, how repeated parallel evolution and adaptation from low-frequency standing variation within a NP producing population occurs is still not clear, especially in light of large effective population sizes for microorganisms. The ‘sampling hypothesis’ requires sampling events to be frequent and always permitted within genomes. Such that how innovation is delivered is still an open question and likely intimately linked to the subsequent questions.
- 2) **Are biosynthetic gene clusters (as multi-part traits) evolving gradually or abruptly?** Duplication and HGT can lead to the recruitment of single or clusters of genes to ‘gene collectives’ (proto-clusters) on which selection can act. This would suggest that gradually or abruptly are both possible if the right genes/proto-clusters/pathways are combined. The presence of hybrid clusters (Jenke-Kodama et al., 2006), natural products serving as precursors for other natural products (Qin et al., 2020), in addition to the emergence of de novo NPs through the same mechanisms. The role of protein-protein interaction and steric hindrance (especially in megasynthases) is likely to play a significant role within the emergence of novel BGCs. The recruitment of promiscuous enzymes into proto-clusters and existing clusters will allow these enzymes to function and supports the ‘sampling hypothesis’. What is less clear in these scenarios is how, newly recruited promiscuous enzymes from primary metabolism may free themselves from the tightly controlled regulation of their activity (post-transcriptional, translational, post-translational) to enable them to fully integrate into a BGC. Perhaps HGT offers a route to avoid this, where post-translationally modified residues are no longer in context, or they are decoupled from existing pathway regulation. The partitioning of primary metabolic enzymes to different functions may in part be achieved by changes in cofactor specificity and enzyme kinetics (Schniete et al., 2018). This is likely to require case by case consideration.

- 3) Which appears first, the biosynthetic gene cluster or the natural product?** Whilst this might seem like an easy question to address it is clear through gene expansion events and the discovery of convergent evolution in NPs that this is not as straightforward as might first appear. This is probably best studied within the antimicrobial NPs, with new examples being increasingly discovered. Convergent evolution can be via evolutionary convergence of pathways to deliver the same molecule (Fischbach, 2009; Grenade, Howe, & Ross, 2021); e.g. beta-lactams (Liras & Martín, 2006), fosfomycin in *Pseudomonas* and *Streptomyces* (Kim et al., 2012), gerumycin and dentigerumycin (Sit et al., 2015), tambjamines in *Pseudoalteromonas* and *Streptomyces* (Grenade et al., 2023) or to deliver highly similar molecules that have converged on the same cellular targets (Fischbach, 2009) (e.g. five structural classes of antibiotic targeting the bacterial ribosome (Hansen et al., 2003), Lipid II (Breukink & Kruijff, 2006), and bacterial cell wall synthesis (Sarkar et al., 2017)). This question requires careful phylogenetic reconstructions and chemical pathway reconstruction and the data and tools to perform these analyses are emerging in the field.
- 4) What is the environmental context of a novel natural product biosynthetic gene cluster/pathway?** NP producing bacteria can be considered as ‘ecotypes’ within complex environments where selective sweeps maintain the structure and function of their genome, morphology, interaction-networks to facilitate survival (Gushgari-Doyle et al., 2022; Prosser, 2022; Wiedenbeck & Cohan, 2011). The emergence of any novel trait is subject to natural selection within a given environment. These traits can exhibit wide taxonomic distribution, can show lineage specific distribution (such as in *Salinispora* Jensen, 2016) or may appear to be lineage independent due to HGT (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019). New traits appear to emerge in lineages of organisms that have shifted their ecological niche or life histories. Likewise, species-niche interactions could also be considered alongside the emergence of novel cell types, such that gene expression is confined to specific aspects of a developmental programme, a phenomenon that is well established in *Streptomyces* bacteria (Hoskisson & Fernández-Martínez, 2018; Hoskisson & Seipke, 2020), *Bacillus* (Maan, Itkin, Malitsky, Friedman, & Kolodkin-Gal, 2022), myxobacteria (Xiao, Wei, Ebright, & Wall, 2011) and fungi (Alberti, Foster, & Bailey, 2017).



9. Placing biosynthetic gene clusters into an ecology–evolution–developmental (eco–evo–devo) biology framework

Use of the term ‘specialised metabolites’ requires context around how the metabolite is specialised in its role. We previously postulated (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019) that knowledge of the ecological role of a NP is required to refer to them as a ‘specialised metabolite’. Here we wish to expand this to account for the wider biology of a producing organism. Linking ecology and evolution in a dynamic manner to explain the driving forces of selection (Eco–evolutionary dynamics) is advancing rapidly and microbiology is at the forefront of emerging paradigms, driven by conceptual and technological advances. The recognition that ecology can drive evolution (eco–evo) has been evident in recent years (Sigmund & Holt, 2021). Yet the role played by NPs of producing organisms can enable them to shape their environment, their populations, communities and physical environments. It is increasingly recognised that evolution can drive ecology (evo to eco) (Bruce & Hoskisson, 2022). Examples from NP producers include symbiosis and environmental interactions (Seipke et al., 2012), where NPs shape interactions with fungus-farming ants (Barke et al., 2010; Cafaro et al., 2011; Seipke et al. 2011; Seipke, Barke, Heavens, Yu, & Hutchings, 2013; Sit et al., 2015), solitary wasps (Kaltenpoth, 2009), marine invertebrates (Selvin, Priya, Kiran, Thangavelu, & Bai, 2009; Taylor, Hill, Piel, Thacker, & Hentschel, 2007) and plants (Kruasuwan et al., 2023; Sousa, de, & Olivares, 2016; Worsley et al., 2020). Niche-specific NPs can enable persistence of an organism in a given environment and this can drive evo–eco dynamics, however, determining the fitness cost of NP production in these is difficult experimentally, especially given the conditions that favour production potentially have costs for primary metabolism and growth (Bayram et al., 2016; Brakhage, 2013; Lind et al., 2015; Osbourn, 2010b; Santamaria et al., 2022). Production of NPs does enable collective traits (nutrient scavenging, microbial warfare and cell signalling) but at the expense of diverting precursors, cofactors and energy away from growth (Santamaria et al., 2022; Schniete et al., 2018). Therefore we need to consider their production within social evolution and kin selection contexts (Santamaria et al., 2022), such as the costs of invoking warfare (Gonzalez, Sabnis, Foster, & Mavridou, 2018; Kirkup & Riley, 2004; Niehus, Oliveira, Li, Fletcher, & Foster, 2021), the role of kin selection

and ‘Greenbeards’ in NP production (Gardner & West, 2010; Lyons, Kraigher, Stefanic, Mandic-Mulec, & Kolter, 2016), and the emergence of evolutionary cheats and thieves, for example siderophore piracy (Bruce, Cooper, Chabas, West, & Griffin, 2017; Kümmerli, 2022). This knowledge will enable us to consider the specialised role NP play in the biology of producers.

To fully develop our view of specialised metabolism we must place NPs in an ecology–evolution–developmental context. Many natural product producers have complex lifecycles, social biology and community interactions that have shaped their biology. The original view in Evo–Devo was that development can shape evolution through setting out possibilities prior to selection (Gilbert, Bosch, & Ledón-Rettig, 2015). However at the heart of Evo–Devo is how plasticity shapes development and selection in the environment. Consider *Streptomyces* bacteria, prolific producers of NPs that exhibit remarkable developmental lifecycles—for decades researchers have sought to identify the trigger for NP synthesis and the trigger for development. The reality is that robustness and plasticity within the genomes of these organisms mean that they can response to multiple cues and triggers, sensing and responding to environmental cues in dynamic ways. Therefore we need to consider how genes, development, plasticity and the environment can shape the evolution of NPs to understand what the ‘specialised’ in ‘specialised metabolism’ is. There is great plasticity in metabolic networks and major differences can manifest even within closely related organisms (Wagner, 2009). This provides the material upon which evolution can work: dynamic genetic circuitry in metabolic pathways enables evolvability (Jiménez, Munteanu, & Sharpe, 2015). There are emerging pictures of how selection can act upon gene expression (Price et al., 2022), however we know that it acts at multiple levels (Roditi, Boyle, & Xavier, 2013). Selection dynamics are also variable, acting on different alleles in a non-continuous manner, given not all gene are always expressed, however there is some evidence that rapid fixation of a beneficial allele may require continuous selection—‘Ohno’s dilemma’ (Berghthorsson, Andersson, & Roth, 2007). This is likely to reflect the dynamic natural environment and that evolution is shaped by numerous evolutionary forces.

We argue that placing NPs within an Eco–Evo–Devo framework can help us understand what ‘specialised’ truly means.

Eco-specialised metabolites that contribute to niche-specific interactions such as those produced by symbiotic species to help maintain the relationship (Barke et al., 2010; Cafaro et al., 2011; Kaltenpoth, 2009;

Kruasuwan et al., 2023; Seipke et al. 2011; Seipke et al., 2013; Selvin et al., 2009; Sit et al., 2015; Sousa et al., 2016; Taylor et al., 2007; Worsley et al., 2020), siderophores where production contributes to competitive nutrient acquisition (Chevrette et al., 2022; Gushgari-Doyle et al., 2022; Kümmerli, 2022; Sathe & Kümmerli, 2020; Vind et al., 2023). Environmental cues for specific nutrient limitations and stresses are also known to activate NP biosynthesis along with sensing and responding to NPs from other organisms (kin and non-kin) (Covington, Xu, & Seyedsayamdost, 2021; Garcie et al., 2016; Gonzalez et al., 2018; Kirkup & Riley, 2004; Niehus et al., 2021; Rigali et al., 2008).

Evo—lineage specific NPs are likely to represent specialised metabolites in their strictest sense, however understanding their exact role and reasons for restricted distribution has been challenging. Pioneering work in the Eco-evo area has been from the Jensen group, where they have shown examples of lineage specific molecules that are globally distributed (Jensen, Williams, Oh, Zeigler, & Fenical, 2006). Where functional redundancy can help shape genomes—the replacement of BGCs at specific chromosomal locations (Ziemert et al., 2014). This could suggest that there are ‘hotspots’ for recruitment, or that evolution constrains the number of clusters to mitigate the evolutionary cost of acquisition (Jensen et al., 2006; Ziemert et al., 2014). There is certainly evidence for genomic islands of NP BGCs (Penn et al., 2009). Evidence of cluster replacement, potentially demonstrating functional competitive exclusion and evidence of selection against superfluous gene cluster retention (Bruns et al., 2018).

Devo—It has long been known that production of NPs and the initiation of development is intimately linked (Alberti et al., 2017; Hoskisson & Fernández-Martínez, 2018; Hoskisson & Seipke, 2020; Maan et al., 2022; Xiao et al., 2011). In *Streptomyces* bacteria there are pleiotropic regulators of both processes that have been extensively studied (Chandra & Chater, 2014; Chater, 1998, 2001; Chater & Horinouchi, 2003; Chater, Biró, Lee, Palmer, & Schrepf, 2010; Heul, van der, Bilyk, McDowall, Seipke, & Wezel, 2018; Hoskisson & Fernández-Martínez, 2018; Hoskisson & Seipke, 2020). Studies of NPs in this context has led to a detailed understanding of how NP expression is coordinated with development (Chandra & Chater, 2014; Chater & Horinouchi, 2003). Roles for specialised metabolites within this developmental framework can be cell-type or tissue specific, or life stage specific such as germination inhibitory and spore associated NPs (Aoki et al., 2011; Kodani et al., 2004; Sigle et al., 2015). More recently division of labour has become an emerging theme in

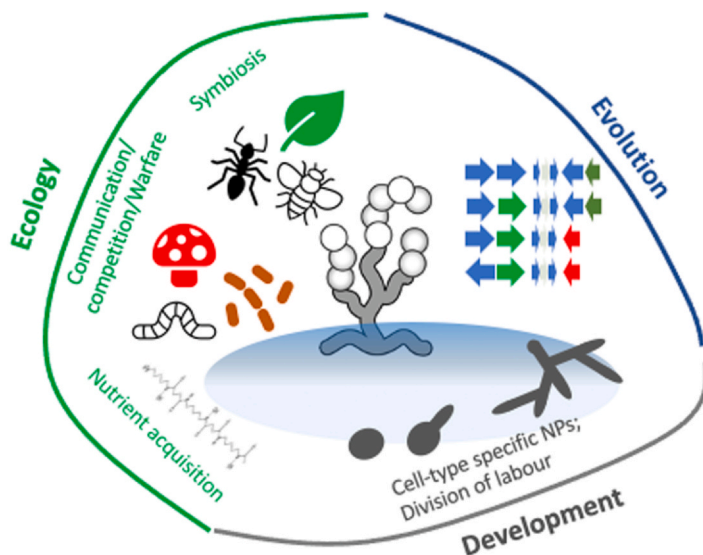


Fig. 4 Placing specialised metabolites into an ecology–evolution–developmental biology (Eco–evo–devo) context. Central to the figure is a NP producing *Streptomyces* colony, surrounded by the known eco–evo–devo roles of specialised metabolites—from driving nutrient acquisition (siderophores), through communication, competition and interspecies warfare and symbiosis (Eco); Evolution of novel NPs clusters (evo); through to developmental specialisation of NP biosynthesis, tissue/cell specific roles and division of labour (devo).

Streptomyces, reconciling cooperative interactions within colonies in terms of metabolism and reproduction (Traxler & Rozen, 2022; Zacharia et al., 2021; Zhang, Shitut, Claushuis, Claessen, & Rozen, 2022), emerging from this is how metabolism may be partitioned to facilitate specific roles that may contribute to specialised metabolite production when organisms are in dynamic developmental transitions (Schniete et al., 2018).

Here we present a framework for considering the roles of NPs and to develop the correct application of ‘specialised metabolites’ as a term rather than it being synonymous with secondary metabolism.

10. Future perspectives

The diversity of natural products is extraordinary, understanding how the diversity is achieved and the mechanisms that govern their evolution can help determine their natural roles and understand the rules by which

BGCs are put together. While huge progress has been made in this area since the discovery and cloning of the first BGCs for NPs over 30 years ago (Hopwood et al., 1985; Smith, Burnham, Edwards, Earl, & Turner, 1990), there has still been a great deal of discussion in the literature around the correct terms to be used for bioactive natural products—primary or integrated, secondary or specialised. Placing these into an eco–evo–devo context can help formalise the roles of these metabolites and consider them in terms of the evolutionary forces that shape them. Progress has also been made in understanding the forces and mechanisms that shape the evolution of biosynthesis that gives rise to the diversity of NPs we see in nature and also in the laboratory.

Looking forward, understanding how metabolism evolves, the mechanisms that drive innovation within precursor supply in addition to NP biosynthesis and its regulation can assist in the development of novel, evolution inspired bioactive NPs, non-natural NPs (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019) and the activation of silent BGCs in the genomes of talented NP producing organisms (Chevrette, Gutiérrez-García et al., 2019; Chevrette, Hoskisson et al., 2019). We are still learning what NP BGCs look like, and current tools for their identification by genome mining rely on us finding known-knowns (Hoskisson & Seipke, 2020). The unknown-unknowns (Hoskisson & Seipke, 2020)—cryptic molecules with cryptic biosynthetic pathways—perhaps represent some of the most exciting discoveries awaiting us in NP research—novel chemical scaffolds with bioactivity that can be exploited for clinical use. How this is accessed remains a challenge—bioinformatic tools are developing all the time to exploit the wealth of data available (Chevrette et al., 2021), understanding the rules of NP biosynthesis will require extensive ecology and evolution studies, and there will be a significant place for experimental evolution (Kawecki et al., 2012) in these studies to fully understand the ‘rules of BGC evolution’ and precursor supply. There is still much to be learned about population level distribution and production of NPs. The emerging picture of populations structure in *Streptomyces* is that there is genetic heterogeneity within cultures with strains unlikely to be monoclonal (Doroghazi & Buckley, 2010; McDonald & Currie, 2017; Tidjani et al., 2019). This is likely to result in phenotypic and chemotypic diversity within populations.

While there is still some way to go in understanding this, progress is being made in using our understanding of the genetic and chemical logic of NP biosynthesis such as refactoring BGCs (Li, MacIntyre, & Brady, 2021),

bio-inspired synthesis (Seipke, 2022; Wang et al., 2022), chemosynthesis and mutasynthesis (Heide, 2009; Weissman, 2007) and combinatorial biosynthesis (Hopwood et al., 1985). The converging of these technological advances along with a deep understanding of the eco–evo–devo landscape of NPs, their chemical and biosynthetic logics will enable engineering biology of de novo BGCs and their products to become a reality.

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